

Synthetic supply networks

Galvin Ng
CSH

Luca Mungo
Oxford INET

Damien Bertrand
EPFL

François Lafond
Oxford INET

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Why synthetic supply networks?

- Why supply networks?
Because COVID-19,
natural disasters, tariffs.
- Affects aggregate
fluctuations, inflation,
systemic risk, resilience
- Heterogeneity matters!

- Data-driven economic ABMs:

nature human behaviour

Article

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The unequal effects of the health–economy trade-off during the COVID-19 pandemic

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Marco Pangallo¹✉, Alberto Aleta², R. Maria del Rio-Chanona³,
Anton Pichler⁴, David Martin-Corral⁵, Matteo Chinazzi^{6,7},
François Lafond⁸, Marco Ajelli⁹, Esteban Moro^{4,9}, Yamil Moreno^{1,10},
Alessandro Vespignani¹¹ & J. Dooyne Farmer¹²

Forecasting Macroeconomic Dynamics using a Calibrated Data-Driven
Agent-based Model*

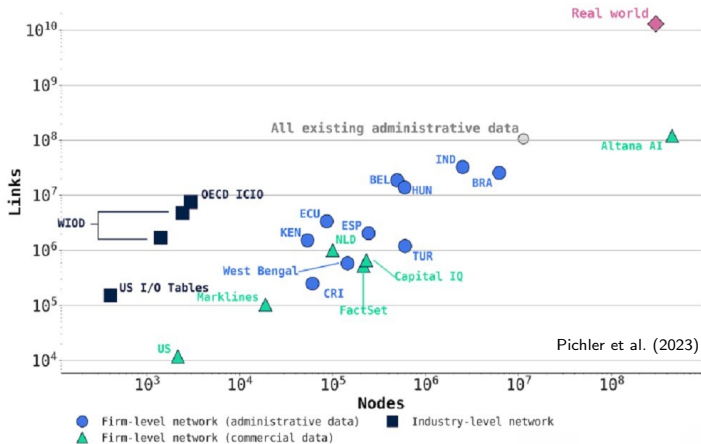
SAMUEL WIESE^{a,b,i}, JAGODA KASZOWSKA-MOJSA^{a,c,d}, JOEL DYER^{a,b}, JOSÉ MORAN^{e,a}, MARCO
PANGALLO^{a,f}, FRANÇOIS LAFOND^{a,g}, JOHN MUELLBAUER^{a,h}, ANISOARA CALINESCU^{a,b}, J. DOYNE
FARMER^{a,e,g,i}

- Synthetic populations of **households**
are increasingly common

What about synthetic populations of **firms**?

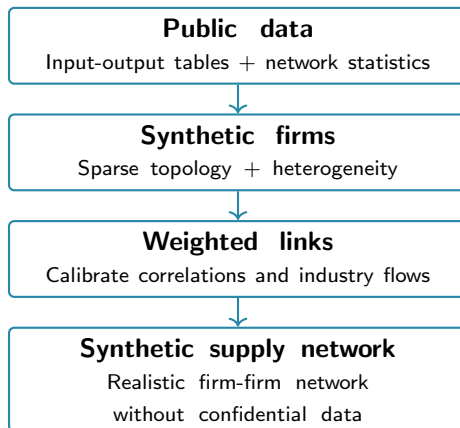
Additional context

Reconstructing the global supply network is an ongoing game



Goal: Develop a synthetic population of firms using only free, publicly available information.

Road map



Why is a representative supply network plausible?

Supply networks exhibit stable statistical regularities

⇒ A representative supply network is plausible

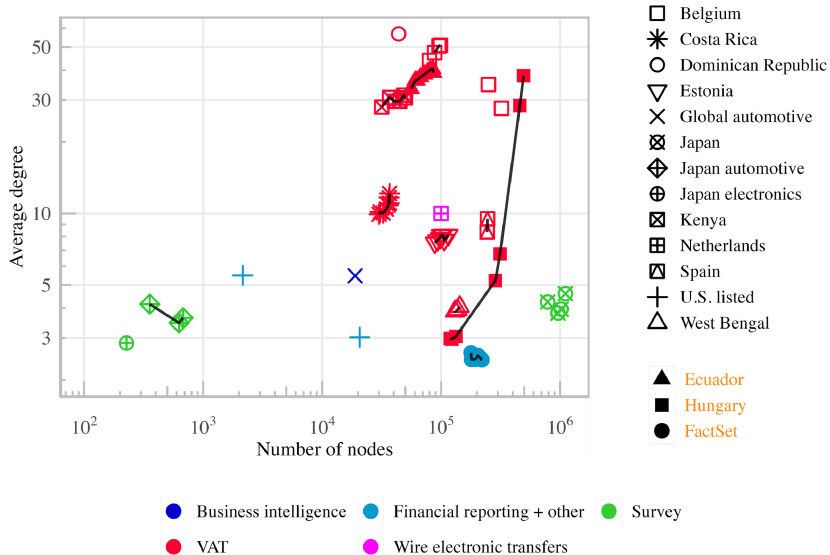
Bacilieri, A., Borsos, A., Astudillo-Estevez, P., Hoefer, M., & Lafond, F. (2026). *Firm-level production networks: what do we (really) know?* *Journal of Economic Dynamics and Control*, 105313.

These regularities strongly constrain network structure

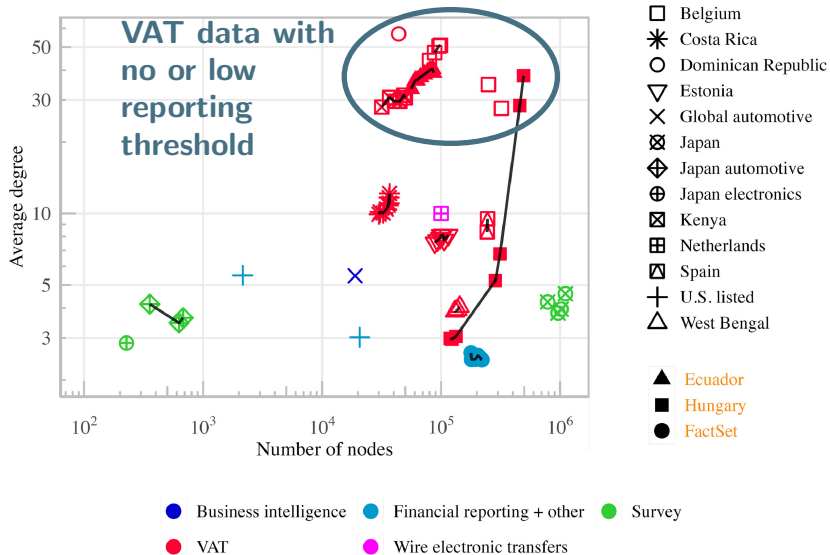
⇒ Only a relatively small set of networks are plausible

Ng, G., Mungo, L., Bertrand, D. & Lafond, F. (in progress). *Synthetic supply networks*.

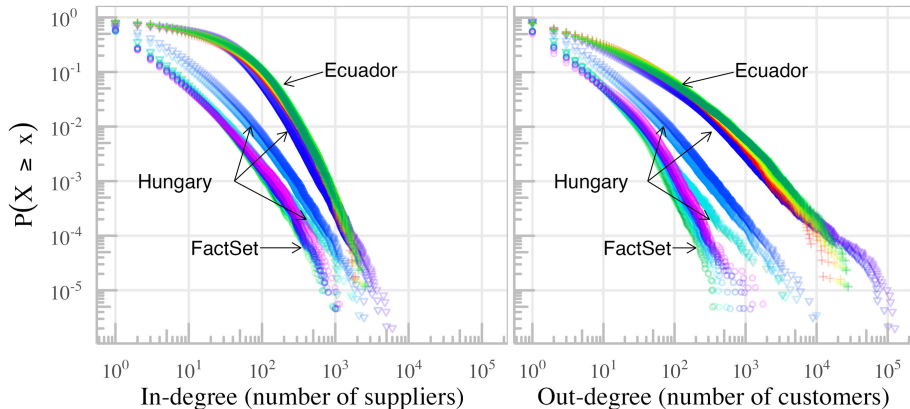
Evidence 1: Different networks, similar average degree



Evidence 1: Different networks, similar average degree



Evidence 2: Different networks, similar tail exponents



Approaches to supply network reconstruction

- Proprietary data providers collect large-scale firm-level supply chain data. Example: Altana AI.
- Machine learning and knowledge graph approaches to infer firm-level and product-level relationships. Examples: Cambridge SCAI-Lab, Mungo et al. (2023); Karbevaska and Hidalgo (2025); Köse et al. (2026).
- Maximum entropy and mechanistic approaches focus on reproducing economic and network properties. Examples: Ialongo et al. (2022); Hurt et al. (2023); Reisch et al. (2026).

Our approach

$$\mathbf{W} = \begin{matrix} & f_1 & \cdots & f_N \\ \begin{matrix} f_1 \\ \vdots \\ f_N \end{matrix} & \begin{pmatrix} W_{11} & \cdots & W_{1N} \\ \vdots & \ddots & \vdots \\ W_{N1} & \cdots & W_{NN} \end{pmatrix} \end{matrix}$$

We want to generate a population of **synthetic** firms $f_1, f_2, \dots, f_N$ within a network that is

- 1 Sparse, directed, weighted
- 2 Match the input-output table chosen by the user
- 3 Capture heterogeneity and descriptive statistics

by strictly using only publicly available information.

Generation procedure

- 1 Generate in- and out-degrees $k_i^{\text{in}}, k_i^{\text{out}}$ from BurrXII distributions¹, to match tail exponents, means and variances of log-degrees
- 2 Noisy rank alignment of k^{in} and k^{out}
- 3 Sample adjacency matrix $\mathbf{A}_{ij}$ from Configuration Model
- 4 Randomly assign firms to industries [► Details](#)
- 5 Sample lognormal fitnesses $f_i^{\text{in}}, f_i^{\text{out}}$
- 6 Compute weights $\mathbf{W}_{ij}^{\text{init}} = \mathbf{A}_{ij} (f_i^{\text{out}})^{\theta_{f,\text{out}}} (f_j^{\text{in}})^{\theta_{f,\text{in}}} (k_i^{\text{out}})^{\theta_{k,\text{out}}} (k_j^{\text{in}})^{\theta_{k,\text{in}}}$, and rescale to match IOT element-wise [► Details](#)

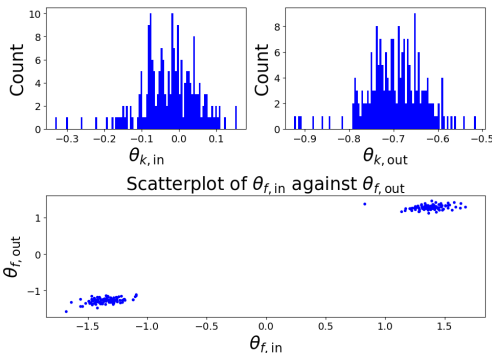
We need to find $\theta_{k,\text{in}}, \theta_{k,\text{out}}, \theta_{f,\text{in}}, \theta_{f,\text{out}}$.

¹Let $X \sim \text{BurrXII}(c, k, \lambda)$, then $\mathbb{P}(X \geq x) = [1 + (\frac{x}{\lambda})^c]^{-k} \sim x^{-ck}$

Calibrating the synthetic network

$$\min_{\theta_{k,\text{in}}, \theta_{k,\text{out}}, \theta_{f,\text{in}}, \theta_{f,\text{out}}} (\mathcal{L}_{\text{strength hill}} + \mathcal{L}_{\text{strength variances}} + \mathcal{L}_{\text{joint distributions}})$$

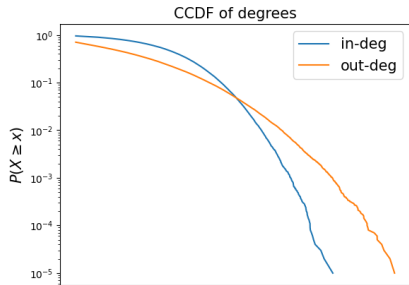
► Details



$$\mathbf{W}_{ij}^{\text{init}} = \mathbf{A}_{ij} \times (f_i^{\text{out}})^{-1.3} (f_j^{\text{in}})^{-1.3} (k_i^{\text{out}})^{-0.7}$$

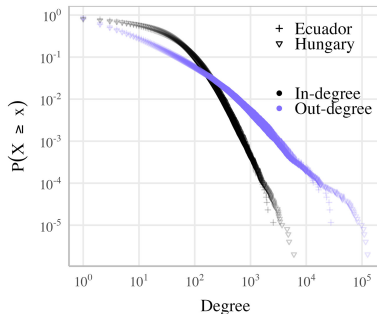
Figure: Distribution of optimal parameters across 200 Optuna studies.

Result: Degree distributions



(a)

Synthetic data: $N = 100k$ firms

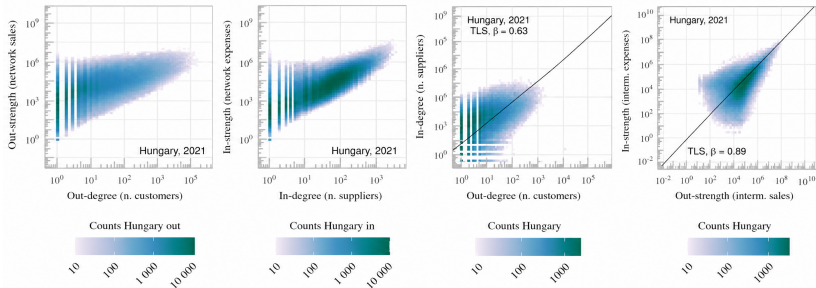


(b)

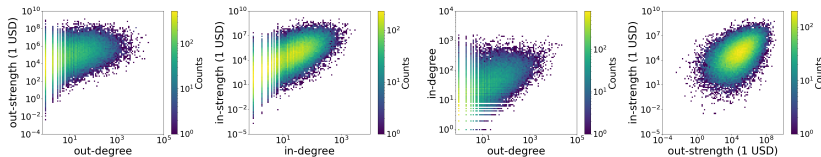
Real data: $N = 100k, 500k$ firms

Figure: Comparison of degree distribution CCDFs.

Result: Joint distributions



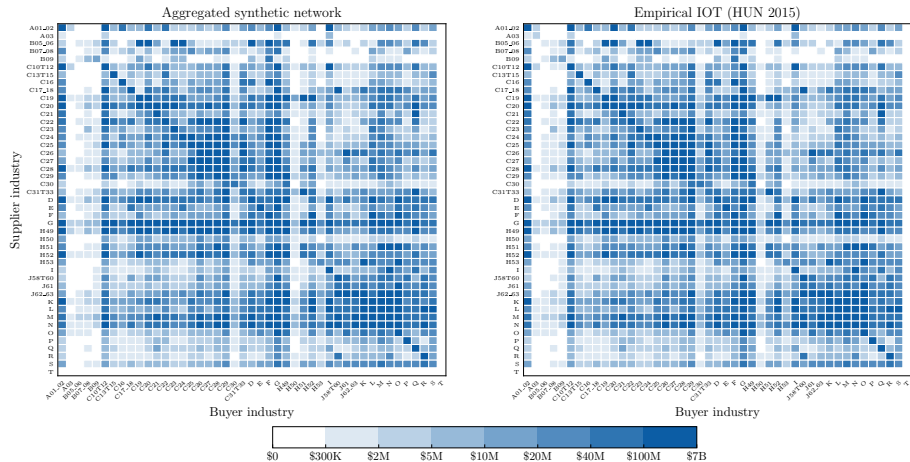
(a) Hungary



(b) Synthetic data

Figure: Comparison of joint distributions:

Result: Industry-level aggregation



Generated for

- Repeated runs: very small standard deviation. [▶ See appendix](#)
- Different countries: Belgium, Hungary, Brunei. [▶ See appendix](#)
- Network sizes: 10k, 50k, 100k, 500k. [▶ See appendix](#)

Takeaways

- 1 We generate synthetic firm-level supply networks using only publicly available data.
- 2 Generated networks reproduce empirical network structure and aggregate economic flows.
- 3 We provide a scalable, reproducible framework to generate synthetic microdata for economic modellers

Working paper: tinyurl.com/ssncef26



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Appendix

Ablation study

	for fixed mean = 40			for mean $\bar{k} = 0.86N^{1/3}$ (rounded to nearest 10)	
	N = 100,000	N = 10,000	N = 1,000	N = 10,000 $\bar{k} = 20$	N = 1,000 $\bar{k} = 10$
Burr with discretization, truncation, and reconciliation					
mean degree	39 (0.917)	33.8 (1.48)	19.6 (1.54)	18.1 (0.933)	7.28 (0.804)
out-deg hill exponent	1.4 (0.0408)	1.67 (0.138)	5.83 (1.96)	1.51 (0.132)	2.53 (0.713)
in-deg hill exponent	2.33 (0.0695)	4.57 (0.389)	24.7 (6.41)	2.9 (0.237)	12.9 (3.78)
out-deg log variance	2.95 (0.0151)	2.92 (0.045)	2.51 (0.105)	2.43 (0.0437)	1.87 (0.111)
in-deg log variance	2.01 (0.0118)	1.93 (0.028)	1.29 (0.0537)	1.68 (0.0294)	1.16 (0.0588)
max out-deg	30,616 (5,939)	3,682 (267)	381 (17.4)	3,277 (512)	330 (50.8)
max in-deg	4,007 (561)	463 (20.8)	73.6 (7.19)	450 (29)	50 (5.87)
Burr with discretization and truncation					
in-deg mean	40 (0.269)	39.9 (0.8)	35.2 (1.66)	20 (0.422)	9.84 (0.581)
in-deg hill exponent	2.31 (0.0687)	2.37 (0.217)	6.45 (1.95)	2.35 (0.227)	2.95 (0.884)
in-deg log variance	2.02 (0.00805)	2.02 (0.0261)	1.93 (0.0744)	1.73 (0.0219)	1.4 (0.0578)
max in-deg	6,728 (3,722)	2,429 (673)	380 (18)	1,434 (630)	247 (69)
Burr with discretization only					
out-deg mean	40 (2.17)	40 (4.83)	40.1 (18)	20 (2.42)	9.96 (3.43)
in-deg mean	40 (0.267)	40 (0.871)	40 (2.7)	20 (0.441)	9.97 (0.665)
out-deg hill exponent	1.39 (0.0416)	1.41 (0.132)	1.67 (0.553)	1.41 (0.131)	1.66 (0.538)
in-deg hill exponent	2.31 (0.0691)	2.35 (0.225)	2.79 (0.929)	2.35 (0.225)	2.76 (0.903)
out-deg log variance	2.95 (0.015)	2.96 (0.0484)	2.96 (0.151)	2.45 (0.0454)	1.99 (0.135)
in-deg log variance	2.02 (0.00808)	2.02 (0.0256)	2.02 (0.0802)	1.73 (0.022)	1.40 (0.0589)
max out-deg	85,028 (171,357)	23,628 (34,725)	6,558 (16,108)	11,994 (17,927)	1,603 (2,777)
max in-deg	6,862 (4,178)	3,123 (2,303)	1,324 (932)	1,525 (998)	332 (226)
Burr					
out-deg mean	40 (3.41)	40 (5.13)	40 (17.7)	20 (2.33)	9.97 (3.80)
in-deg mean	40 (0.277)	40 (0.855)	40 (2.74)	20 (0.43)	9.99 (0.685)
out-deg hill exponent	1.39 (0.0409)	1.4 (0.132)	1.59 (0.529)	1.41 (0.132)	1.6 (0.543)
in-deg hill exponent	2.31 (0.0681)	2.34 (0.221)	2.66 (0.911)	2.33 (0.226)	2.64 (0.91)
out-deg log variance	10 (0.0611)	10 (0.195)	10 (0.612)	10 (0.198)	10 (0.617)
in-deg log variance	3 (0.0182)	3 (0.0583)	3 (0.183)	3 (0.0587)	3 (0.185)
max out-deg	84,604 (312,946)	23,879 (38,485)	6,519 (15,630)	11,901 (16,318)	1,581 (3,144)
max in-deg	6,899 (4,097)	3,108 (2,099)	1,337 (987)	1,563 (1,061)	334 (238)

Selecting parameters α, ν for BurrXII out-degrees

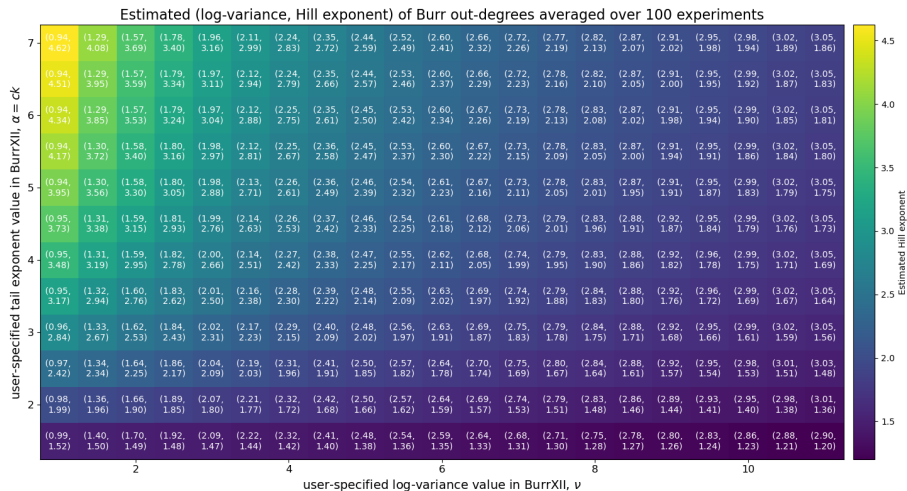


Figure: We choose $\alpha = 2$, $\nu = 10$.

Selecting parameters α, ν for BurrXII in-degrees

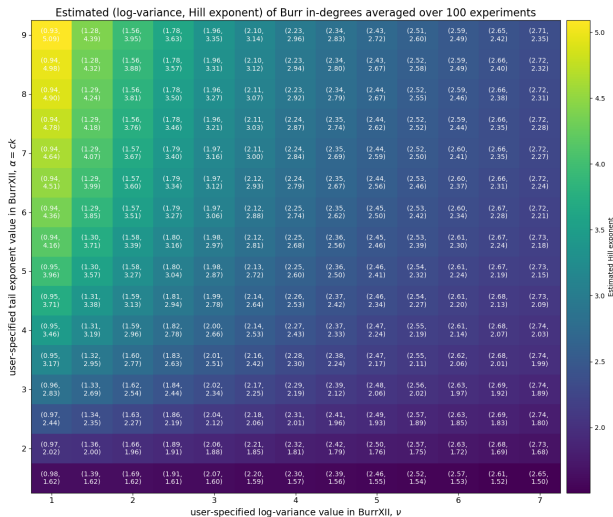


Figure: We choose $\alpha = 3$, $\nu = 3$.

Why use BurrXII to sample degrees?

Denote a Burr random variable with strictly positive parameters (c, k, λ) by $X \sim \text{BurrXII}(c, k, \lambda)$, with Cumulative Distribution Function (CDF)

$$F(x) = 1 - \left[1 + \left(\frac{x}{\lambda} \right)^c \right]^{-k}.$$

Observe that as $x \rightarrow \infty$, $1 - F(x) \sim x^{-ck}$, so the asymptotic tail exponent is

$$\alpha = ck.$$

Moreover, its expectation is

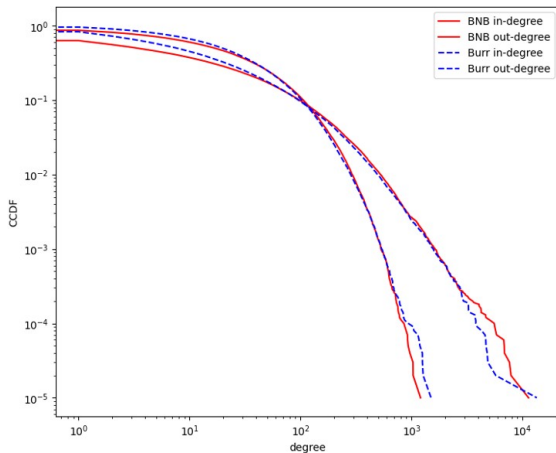
$$\mathbb{E}[X] = \lambda k B\left(k - \frac{1}{c}, 1 + \frac{1}{c}\right),$$

for $B(\cdot, \cdot)$ is the Beta function. Furthermore, the log-variance is given by

$$\text{Var}(\ln X) = \frac{\frac{\pi^2}{6} + \psi_1(k)}{c^2},$$

where $\psi_1(\cdot)$ is trigamma function.

Burr degrees vs Beta Negative Binomial degrees



Metric	BNB	Burr
LWCC	86557	95593
Avg path length	2.88	2.95
Degree assortativity	-0.018	-0.015
Reciprocity	0.003	0.002
Avg clustering	0.025	0.017
Global clustering	0.014	0.010

Allocating firms to industries

For N firms, M industries,

- 1 Determine the preliminary industry count for industry $s = 1, \dots, N$,

$$\tilde{M}_s = \text{round} \left(N \frac{\sum_{n=1}^M \text{IoT}_{sn}}{\sum_{m=1}^M \sum_{n=1}^M \text{IoT}_{mn}} \right)$$

- 2 Due to rounding error, adjust $(\tilde{M}_s)_{s=1}^M$ to get $(M_s)_{s=1}^M$
 - Add one count to, or subtract one count from, a uniformly randomly selected industry among those with positive current counts repeatedly, until $\sum_s M_s = N$.
- 3 Allocate firms into industries linearly, i.e. first M_1 firms into industry 1, next M_2 firms into industry 2, $\dots$

Aggregating firm-level network

For N firms, M industries, we have

$$\mathbf{W}_{ij}^{\text{init}} = \mathbf{A}_{ij} (f_i^{\text{out}})^{\theta_{f,\text{out}}} (f_j^{\text{in}})^{\theta_{f,\text{in}}} (k_i^{\text{out}})^{\theta_{k,\text{out}}} (k_j^{\text{in}})^{\theta_{k,\text{in}}},$$

then the rescaled weighted network is given by

$$\mathbf{W}_{ij} = \frac{\text{IOT}_{g_i g_j}}{\sum_{f \in g_i} \sum_{h \in g_j} \mathbf{W}_{fh}^{\text{init}}} \mathbf{W}_{ij}^{\text{init}}.$$

Equivalently, define the $M \times N$ aggregator matrix $\mathbf{S}$ by

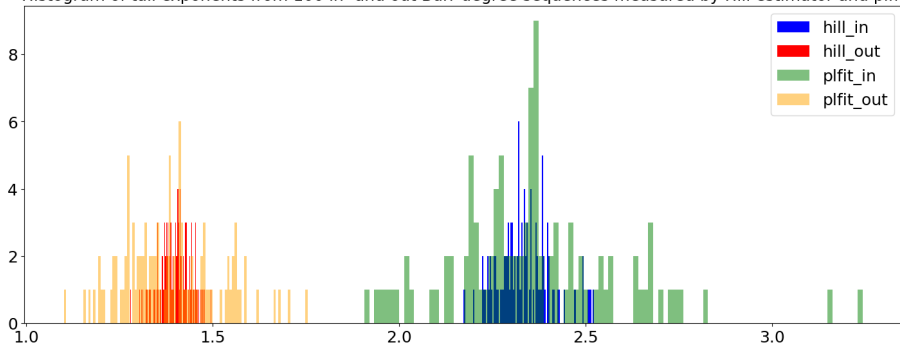
$$\mathbf{S}_{uj} = \begin{cases} 1 & \text{if firm } j \text{ belongs to industry } u \\ 0 & \text{otherwise} \end{cases},$$

then the rescaled weighted network is given by

$$\mathbf{W} = \mathbf{S}^\top \left(\text{IOT} \oslash \left[\mathbf{S} \mathbf{W}^{\text{init}} \mathbf{S}^\top \right] \right) \mathbf{S} \odot \mathbf{W}^{\text{init}}.$$

Choosing tail probabilities for Hill estimator

Histogram of tail exponents from 100 in- and out Burr degree sequences measured by Hill estimator and plfit



	mean	median	std	min	max
hill_in	2.33	2.33	0.0715	2.17	2.52
hill_out	1.4	1.4	0.0419	1.28	1.5
plfit_in	2.36	2.35	0.228	1.91	3.24
plfit_out	1.39	1.38	0.129	1.1	1.75

$$\text{Hill estimator} = \begin{cases} \frac{n}{\sum_{i=1}^n \ln \frac{x_i}{x_{\min}}} & \text{if continuous,} \\ \frac{n}{\sum_{i=1}^n \ln \frac{x_i}{x_{\min} + \frac{1}{2}}} & \text{if discrete} \end{cases}$$

Full loss function

$$\begin{aligned}\text{Loss} = & \mathcal{D}(\text{hill}(\text{in-str})) + \mathcal{D}(\text{hill}(\text{out-str})) \\ & + \mathcal{D}(\text{Corr}(\text{out-str}, \text{out-deg})) + \mathcal{D}(\text{Corr}(\text{in-str}, \text{in-deg})) \\ & + \mathcal{D}(\text{OLS}(\text{in-str}, \text{in-deg})) + \mathcal{D}(\text{OLS}(\text{in-deg}, \text{in-str})) \\ & + \mathcal{D}(\text{OLS}(\text{out-deg}, \text{out-str})) + \mathcal{D}(\text{OLS}(\text{out-str}, \text{out-deg})) \\ & + \mathcal{D}(\text{TLS}(\text{in-str}, \text{out-str})) \\ & + 10^{-3} \times \mathcal{D}(\text{Var}(\text{in-str})) + 10^{-3} \times \mathcal{D}(\text{Var}(\text{out-str})),\end{aligned}$$

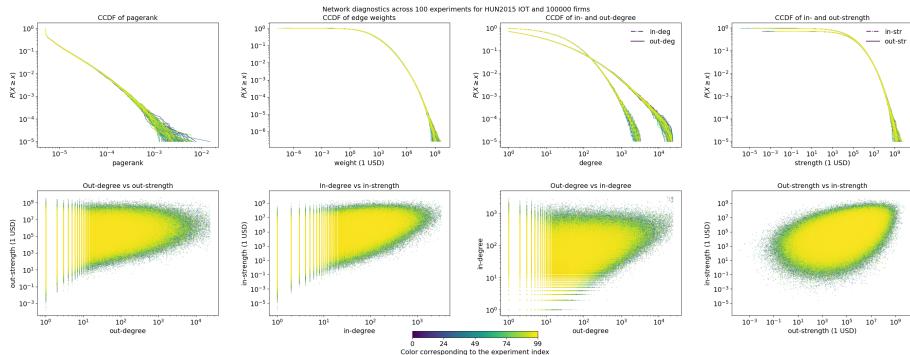
where, for any statistic q , $\mathcal{D}(q) = [\hat{q}(\boldsymbol{\theta}) - q^*]^2$ is the squared discrepancy between its value in the synthetic network, $\hat{q}(\boldsymbol{\theta})$, and its empirical target, q^* .

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Full result

Metric	Value	Empirical	Metric	Value	Empirical
Number of nodes	100,000	100,000	Hill in-degree	2.54	2.5
Number of edges	3,849,303	4,000,000	Hill out-degree	1.48	1.5
Share of $k^{\text{out}} = 0$ (%)	26.20		Hill in-strength	1.05	1
Share of $k^{\text{in}} = 0$ (%)	0		Hill out-strength	1.08	1
Mean degree	38.49	40	Hill influence	1.44	[1.2, 1.3]
Max in-degree	3,065		Hill weights	1.08	[1.1, 1.2]
Max out-degree	21,088		Corr: $k^{\text{in}} \sim k^{\text{out}}$	0.549	0.55
Mean log in-degree	3.27	3	Corr: $s^{\text{in}} \sim s^{\text{out}}$	0.5	0.5
Var log in-degree	1.32	2	Corr: $s^{\text{out}} \sim k^{\text{out}}$	0.438	0.5
Mean log out-degree	2.19	2	Corr: $s^{\text{in}} \sim k^{\text{in}}$	0.585	0.75
Var log out-degree	2.97	3	OLS: $s^{\text{out}} \sim k^{\text{out}}$	0.77	0.76
Mean log in-strength (1USD)	10.88	10	OLS: $k^{\text{out}} \sim s^{\text{out}}$	0.249	0.33
Var log in-strength	8.97	9	$R^2(\text{out})$	0.192	0.25
Mean log out-strength (1USD)	10.92	10	OLS: $s^{\text{in}} \sim k^{\text{in}}$	1.528	1.4
Var log out-strength	9.19	8	OLS: $k^{\text{in}} \sim s^{\text{in}}$	0.224	0.4
Giant component size	95,572		$R^2(\text{in})$	0.342	0.56
Avg path length	2.76	3	TLS: $s^{\text{in}} \sim s^{\text{out}}$	0.976	1
Degree assortativity	-0.062	[-0.2, -0.01]	TLS: $k^{\text{in}} \sim k^{\text{out}}$	0.496	0.7
Reciprocity	0.006	[0.03, 0.05]			
Avg clustering	0.084	[0.19, 0.28]			
Global clustering	0.022	low			

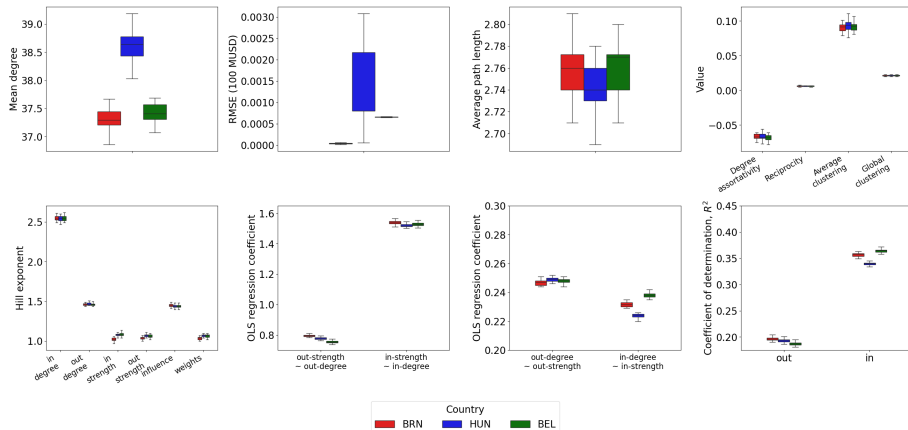
Robustness checks: Repeated runs



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Robustness checks: Different countries

Selected network properties across countries



Robustness checks: Different network sizes

Selected network properties for HUN2015 for different network sizes, with fixed mean degree 40

